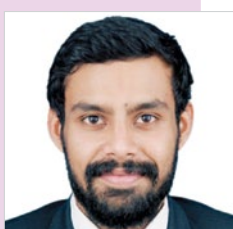


Digitise At Source, Store In The Cloud—THE DAQ MANTRA



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More for less' seems to be the trend for data acquisition (DAQ) and logging systems. "In the recent years, we have witnessed a migration to more affordable and compact products for data-logging applications," says Shayan Ushani, business developer, Analog Arts. "The new products offer more functionalities and recording length, and are most often supported by powerful application software," he adds.

The enabling factors

Progress in analogue signal processing and signal detection is a significant reason for development of more powerful data-logging products, feels Ushani. No wonder then that mixed-signal DAQ and data-processing capabilities are features integrated within these devices. "In the instrumentation area, the most important aspect is how you convert physical signals into electrical signals," says Vinod Mathews, founder and CEO, Captronics Systems Pvt Ltd.

Ushani reiterates, "Data-logging applications are usually concentrated around monitoring the output of a sensor. The sensor can be an ordinary thermocouple sensor or a sophisticated earthquake detector." Powerful application software, easy-to-adapt equipment and automation seem to be the dominating features in the latest range of

DAQ and data-logging products.

Mathews feels that the change is not in the sensor, its accuracy or calibration, but in the treatment of signals. He says, "Today, you are digitising at source and sending the digital signal; so you are not worried about interference by motors and pumps, or other electromagnetic interference."

Smartly exploiting the digital signal

"The idea is how to get more data into a smaller frequency range," continues Mathews. He explains that the allotted spectrum is fixed, and adding more data to it will only introduce high density and very low fidelity. Some of the traditional methods for wideband communication are amplitude-shift keying, phase-shift keying, amplitude- and phase-shift keying, frequency-shift keying, and analogue modulation techniques like amplitude modulation and frequency modulation. "Today, we are moving towards orthogonal frequency-division multiplexing (OFDM)," he adds.

Mathews feels that the main advantage offered by OFDM is the ability to encode digital data on multiple carrier frequencies. "This is already being used in 4G, wireless and regular radio frequency applications, during acquisition. Basically, OFDM has multiple carriers spread over a large area and low symbol rate to get higher data rate than that carried by a single signal. In other words, we get a wider band," he elaborates.



Fig. 1: USB data logger from Labjack - Digit TLH
(Image courtesy: <https://labjack.com>)

What kind of data do you need

Take the case of an automobile. The demand is to know when, where and how you make the measurement. It is not to learn if the car is functioning right, but to get statistical information to improve efficiency of the plant, and not just fix an issue in a particular unit. Today, if you go and give your car at the service station, they know everything from the vendor and date of manufacture, to who fixed what problem at your last service. This is possible because data has been acquired at every point, right from manufacturing and assembly to outward quality check from the tier 1 supplier and inward quality of the original equipment manufacturer. Information is being captured in terms of what is causing more problems, which user has a problem, what kind of defects are possible and when those defects can happen. Necessary analysis is then performed. Authentication and password protection with biometric entry is the security criteria.

—Based on interaction with Vinod Mathews, founder CEO, Captronics Systems Pvt Ltd